

Transfermarkt Market Values vs EA FC Ratings

Machine Learning Analysis of Player Valuation and Rating Systems

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[GitHub Repository](#)

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Abstract

This study investigates the statistical relationship between player market values on Transfermarkt and overall ratings in EA Sports FC. Using a custom-built dataset of approximately 15,000 matched players across 33 leagues, three modeling approaches were employed: a rule-based baseline heuristic, linear regression, and XGBoost regression, with model interpretability provided by SHAP analysis. The XGBoost model achieved an R^2 of 0.900 and RMSE of 2.17 rating points, indicating that market-based and contextual features explain a substantial portion of rating variance. SHAP analysis revealed age as the most influential predictor, followed by peak and previous market values. Key structural findings include: (1) an "aging paradox" where market value declines with age while EA FC ratings increase due to experience and reputation; (2) league-based adjustments reflecting prestige effects (La Liga +0.91) and market inflation penalties (MLS -0.95); and (3) positional biases indicating that goalkeepers are systematically undervalued in the transfer market relative to their ratings. These findings suggest that while Transfermarkt values and EA FC ratings measure fundamentally different constructs—economic investment versus current ability with reputation—they share a strong common statistical signal, making market-based variables reliable proxies for player quality assessment in large-scale analyses.

Introduction

Understanding the relationship between observable player characteristics and their performance evaluations in popular football datasets has long been a subject of interest. As a football enthusiast, the author frequently observed that player market values reported on Transfermarkt appeared notably consistent with the overall ratings assigned in EA Sports FC (formerly FIFA). This observation raises a compelling question: is this similarity merely coincidental, or does a measurable statistical connection exist between football market valuation and in-game player ratings?

Accurately assessing a football player's ability typically requires far more information than market value alone. Professional scouting reports, technical skills, tactical awareness, physical attributes, and detailed performance statistics all contribute meaningfully to any comprehensive evaluation. Consequently, it would be unrealistic to expect market values to fully explain player ratings.

Nevertheless, the apparent correspondence between these two widely referenced data sources makes the topic worthy of systematic investigation. This study explores this connection by integrating player market data from Transfermarkt with overall ratings from EA Sports FC. The objective extends beyond constructing a predictive model; it aims to quantify the statistical relationship between these two systems and assess the strength of their association. To achieve this, player records from both sources were collected, matched, and analyzed using machine learning models complemented by SHAP-based explainability techniques. The findings offer insights into the degree of alignment between football market valuations and EA FC ratings, as well as the relative contribution of various player characteristics in explaining rating variation.

Baseline Rating Approximation

Before applying machine learning models, a simple rule-based baseline was constructed to evaluate how effectively EA FC overall ratings could be approximated using only market value and player age.

The primary assumption underlying this baseline is that transfer market values inherently aggregate substantial football-related information, including player performance, reputation, and perceived ability. However, market values are also significantly influenced by age, as younger players often command premium valuations due to their long-term potential. To partially correct for this effect, an age-adjusted scaling was applied. The adjustment is centered around an assumed peak performance age of approximately 28

years and introduces a non-linear age-dependent multiplier. The adjusted value is defined as:

$$BaseValue = \left(\frac{MarketValue}{1000} \right) \times 1.27^{(Age-28)}$$

Because the exponent depends on age, the scaling factor varies across different age groups in an exponential manner. Players younger than 28 generally receive a scaling factor below 1, while those older than 28 receive progressively larger multipliers.

After computing the adjusted value, a threshold-based rating function was constructed to map BaseValue into approximate EA FC overall ratings. Each interval was empirically aligned with observed rating distributions in the dataset. Representative examples of this mapping include:

- BaseValue $\geq$ 120,000 $\rightarrow$ Rating $\approx$ 92
- BaseValue $\geq$ 110,000 $\rightarrow$ Rating $\approx$ 91
- BaseValue $\geq$ 95,000 $\rightarrow$ Rating $\approx$ 90
- BaseValue $\geq$ 80,000 $\rightarrow$ Rating $\approx$ 89

The thresholds were calibrated empirically on a training subset by minimizing prediction error relative to observed EA FC ratings.

It should be noted that the relationship between BaseValue and ratings is not uniform across player roles. Goalkeepers, in particular, tend to exhibit systematically different market value distributions compared to outfield players with similar overall ability. Consequently, slight threshold adjustments were required to better approximate ratings for goalkeepers and account for this structural difference in the transfer market.

Despite its simplicity, this baseline achieved an R^2 score of approximately 0.82 on a held-out test set. This suggests that a substantial portion of EA FC rating variation can already be explained using age-adjusted market value alone, providing a reference point for evaluating more complex models in subsequent sections.

Data Collection and Dataset Construction

Following the establishment of the initial baseline models, the project's next phase focused on constructing a large-scale and reliable dataset by integrating player information from EA Sports FC 26 and Transfermarkt.

Extracting EA FC Player Data

The EA FC dataset was sourced from the internal database file `fifa_ng_db.db`. This database is highly normalized, with most player attributes stored using numerical identifiers, while descriptive labels—such as league names, teams, and positions—are stored in separate relational tables. To reconstruct complete player profiles, multiple tables were merged through relational joins. Following integration, the final EA FC dataset included the following attributes:

- Player name
- Date of birth
- Age
- Nationality
- Team
- League
- Position
- Jersey number
- Overall rating

From approximately 26,000 total players in EA FC 26, around 18,000 male players were extracted for analysis.

Collecting Transfermarkt Data

The second data source was Transfermarkt, which provides detailed information on player market values and performance statistics. Initially, a custom web scraping pipeline was developed using `requests` and `BeautifulSoup`. However, this approach proved limited due to inconsistent page structures across leagues and incomplete player listings. To improve reliability and scalability, the `tmkt` Python library was adopted, enabling structured and systematic extraction of player data across the 33 leagues included in EA FC 26.

The extracted features included:

- Player identity attributes (name, age, date of birth, nationality)
- Team and league information
- Playing position and jersey number

- Performance statistics (matches played, minutes, goals, assists, clean sheets, goals conceded)
- Market value indicators (current, previous, and peak market value)

In total, approximately 17,000 player records were collected from Transfermarkt.

Player Matching Across Data Sources

A key challenge in this study was aligning player records between EA FC and Transfermarkt, as no shared unique identifier exists. Moreover, player names are often inconsistent across sources due to formatting differences, spelling variations, and localization. To address this, a multi-stage deterministic matching strategy was implemented using stable player attributes:

- Date of birth
- Nationality
- Team
- Jersey number

Stage 1 — Exact Match (4 Conditions)

Players were first matched using all four attributes simultaneously: birthdate + nationality + team + jersey number. This stage produced the highest-confidence matches.

Stage 2 — Relaxed Matching (3 Conditions)

Unmatched players were subsequently processed using combinations of three attributes:

- birthdate + nationality + team
- birthdate + nationality + jersey number
- birthdate + team + jersey number
- nationality + team + jersey number

This step increased coverage while maintaining strong identity constraints.

Stage 3 — Partial Matching (2 Conditions)

Finally, a limited set of two-condition matches was applied to handle inconsistencies in the raw data:

- team + birthdate

- nationality + birthdate

Each match was assigned a match_type label to track confidence levels and enable later filtering or robustness checks.

Final Dataset

From approximately 17,000 Transfermarkt records, the matching pipeline successfully aligned around 15,000 high-confidence matched players. The final dataset includes the following unified attributes:

- Age
- Position
- Nationality
- League
- Team
- EA FC overall rating
- Current market value
- Previous market value
- Peak market value

This integrated dataset forms the foundation for all subsequent statistical analysis, machine learning modeling, and SHAP-based explainability evaluations.

Linear Regression Baseline

After constructing the unified dataset, a Linear Regression model was employed as an initial interpretable benchmark. To increase modeling flexibility, the next stage introduces linear regression as a statistical benchmark before proceeding to more flexible non-linear approaches.

Data Preprocessing

Before training the model, records with missing values in key variables were removed, including:

- Age

- Market value
- EA FC overall rating
- Position
- League

Players with non-positive market values were excluded. Additionally, low-confidence matches—specifically the "nationality + team only" match category—were filtered out to ensure higher data reliability.

Feature Engineering

The linear model was built on a compact but economically meaningful feature set:

- **Age²** — to capture non-linear career progression effects
- **log(1 + market value)** — to reduce skewness and model diminishing returns in valuation
- **log(1 + previous market value)** — to incorporate recent valuation trajectory
- **log(1 + peak market value)** — to reflect historical maximum perceived potential
- **Position (categorical)** — to account for structural rating differences (e.g., goalkeepers vs. outfield players)
- **League (categorical)** — to capture competitive environment effects

Categorical variables were encoded using one-hot encoding.

Model Training

The dataset was split using an 85/15 train-test holdout strategy. A standard Ordinary Least Squares (OLS) Linear Regression model was trained on the engineered feature set. No target leakage was introduced, as all features were strictly derived from Transfermarkt attributes and player metadata, independent of EA FC ratings.

Results

On the held-out test set, the model achieved:

- **$R^2 = 0.872$**
- **RMSE = 2.46** rating points

When predictions were rounded to the nearest integer to reflect the discrete nature of EA FC ratings:

- **Rounded $R^2 = 0.871$**

Interpretation

These findings indicate that a substantial portion of EA FC rating variance is statistically associated with Transfermarkt market-based variables, driven by current valuation, historical signals, age dynamics, positional structure, and league environment. The remaining unexplained variance suggests the presence of non-linear interactions and higher-order effects, motivating the use of tree-based ensemble models in subsequent analysis.

Machine Learning Framework: XGBoost Regression

To further investigate the relationship between Transfermarkt market-based variables and EA FC overall ratings, an XGBoost (Extreme Gradient Boosting) regression model was employed. Rather than focusing solely on prediction, the primary objective was to evaluate how much of the variation in EA FC ratings is statistically associated with player valuation and contextual information available from Transfermarkt. XGBoost was selected due to its strong performance on structured tabular data and its capacity to capture complex non-linear relationships that may not be adequately represented by traditional linear models.

1. Data Preparation and Preprocessing

A data cleaning pipeline was applied to improve data quality and consistency prior to model training:

- **Filtering:** Non-informative identifiers and matching-related fields were removed. Records with missing values in essential variables or non-positive market values were excluded.
- **Match Quality Control:** To reduce potential noise introduced during dataset synchronization, low-confidence player matches were excluded from the analysis.
- **Categorical Encoding:** Player position and league information were encoded using one-hot encoding, allowing categorical context to be incorporated into the model.

2. Feature Engineering and Selection

Several transformations were introduced to better represent player valuation and career-related effects:

- **Logarithmic Market Value Features:** The natural logarithm was applied to Current, Previous, and Peak Market Values, reducing skewness in player valuations and helping stabilize feature distributions.
- **Age Effects:** An Age^2 term was included to allow the model to capture potential non-linear age-related patterns, such as the peak and decline phases of a career.

The final feature set—comprising Age^2 , log-transformed market values, Position, and League—provides both economic and contextual information about each player.

3. Model Configuration and Training

The XGBoost regressor was configured to balance model complexity and generalization performance:

- **Hyperparameters:** 500 estimators, maximum tree depth of 5, learning rate of 0.05
- **Regularization:** Subsample and column sampling ratios of 0.80
- **Evaluation:** Results were measured using an 80/20 train-test split to ensure generalizability to previously unseen players

4. Rating Calibration

Because EA FC ratings are discrete integer values while regression models generate continuous outputs, a calibration procedure was applied following prediction. The calibration optimized a global prediction shift and a rounding threshold for integer conversion, improving alignment with observed EA FC ratings while preserving the underlying continuous relationships learned by the model.

5. Results

The model achieved the following performance on the held-out test set:

- **$R^2 = 0.9002$**
- **RMSE = 2.166** rating points

This performance suggests that a large proportion of the variation observed in EA FC ratings is statistically explained by market-based and contextual variables, indicating a strong underlying statistical relationship between economic valuation and in-game ratings.

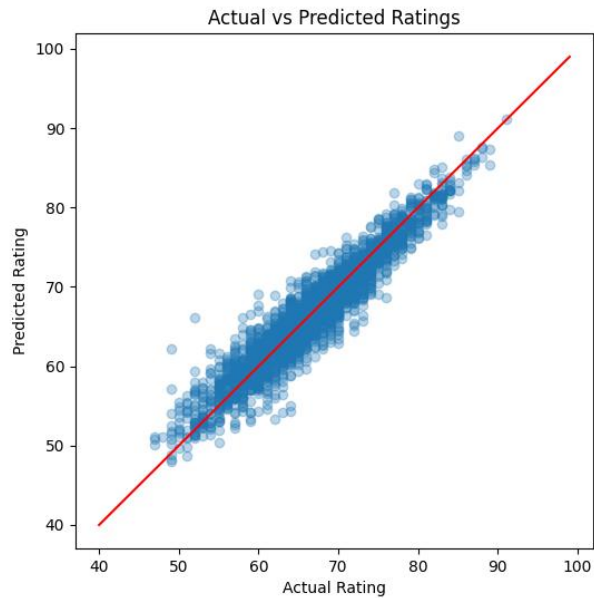


Figure - 1

Actual vs. Predicted Ratings

The scatter plot demonstrates a strong relationship between observed EA FC ratings and model estimates, with most observations located close to the identity line across the entire rating spectrum.

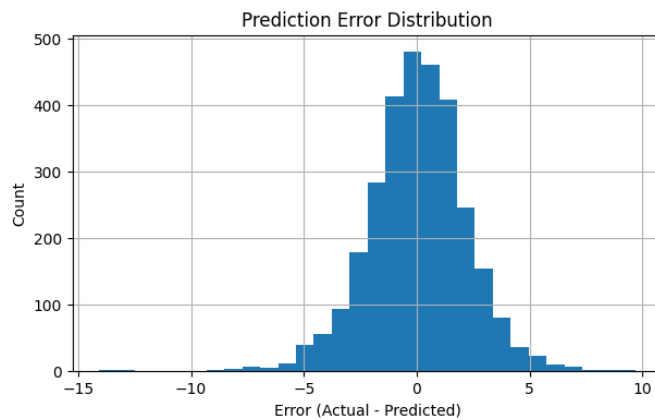


Figure - 2

Prediction Error Distribution

Prediction errors are centered near zero, indicating limited systematic bias. The narrow spread of errors suggests that Transfermarkt-derived variables capture a substantial share of the information associated with EA FC overall ratings.

6. Interpretation

The results provide evidence that player market valuation, together with age, league environment, and positional context, is strongly associated with EA FC overall ratings. While market value alone does not fully determine ratings, the high explanatory power ($R^2 \approx 0.90$) suggests that it contains a substantial predictive signal aligned with how player quality is assessed within EA FC.

Model Explainability Using SHAP

Although the XGBoost model demonstrates strong predictive performance, understanding which variables drive these predictions is essential. To interpret the model, SHAP (SHapley Additive exPlanations) was employed. SHAP provides a game-theoretic approach that measures the contribution of each feature to the predicted rating, enabling both global interpretation of feature importance and local explanations of individual variable influence.

Global Feature Importance

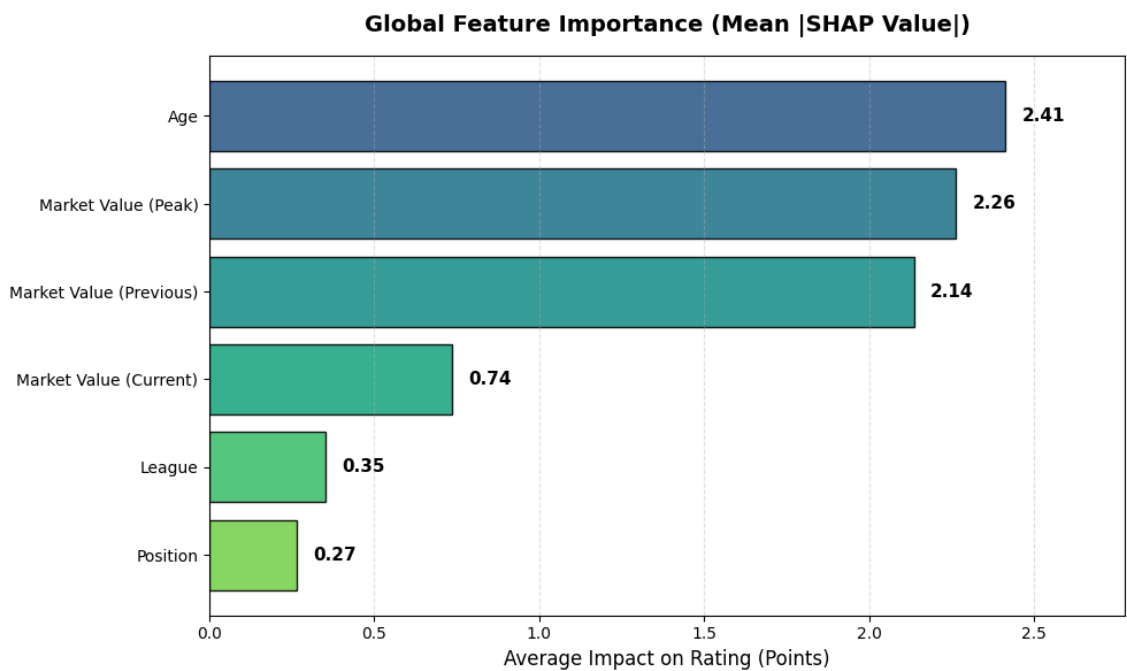


Figure - 3

SHAP Feature Importance

Figure 3 illustrates the global feature importance derived from mean absolute SHAP values. The results reveal that a small group of variables dominates the model's explanatory structure.

Among the most influential predictors, age-related effects emerge as a major contributor, suggesting that career stage is a key determinant of predicted EA FC ratings. Following age, peak market value appears as one of the strongest predictors. Peak value reflects the highest valuation a player has achieved during their career, which often corresponds to the period of maximum performance. Consequently, the model captures peak market value as an indicator of historical peak ability.

Another highly influential feature is previous market value. The strong importance of this variable does not necessarily imply that it is intrinsically more important than current market value. Rather, previous market value often contains overlapping information with current valuation while also reflecting the player's recent market trajectory. In many cases, previous value helps the model capture both the current economic perception of a player and their valuation in the preceding period. Additionally, EA FC overall ratings tend to change more gradually from season to season compared to the more volatile market value fluctuations observed on Transfermarkt. Due to this difference in dynamics, previous market value may align more closely with the valuation context used when new season ratings are assigned.

Importantly, the three market-related variables—current value, previous value, and peak value—together form the economic foundation that largely determines the baseline rating predicted by the model. Following these variables, league context appears as the next most relevant feature group. While its contribution is smaller than the market value variables, league information still improves predictive accuracy by capturing differences in competitive environments. Finally, player position shows the lowest relative importance, though it remains useful for refining predictions and contributes to the model's final precision.

Feature Contribution Patterns

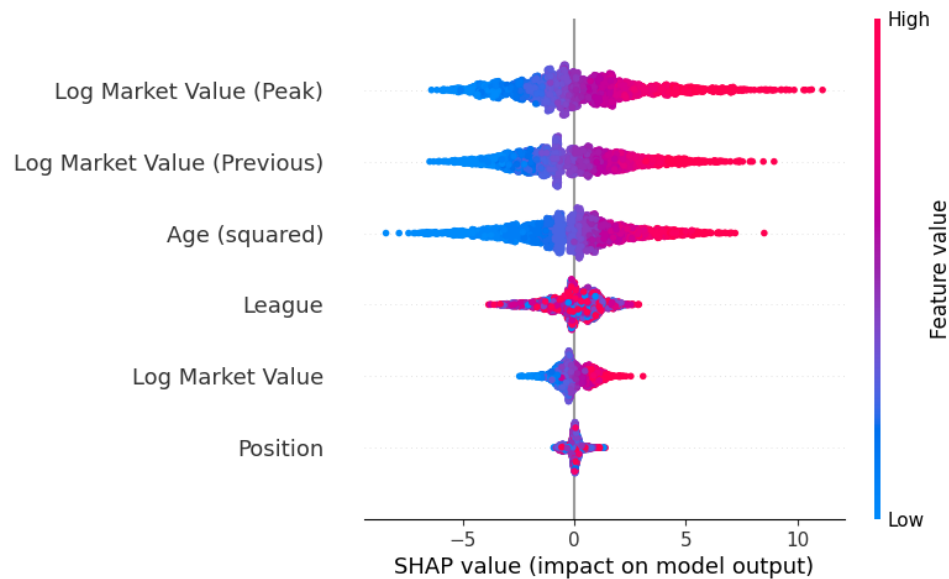


Figure - 4

SHAP Value Distribution

While Figure 4 identifies which features are most influential overall, it does not illustrate how these variables affect predictions. To better understand the direction and distribution of these effects, Figure 4 presents the SHAP value distribution across the dataset. In this plot, each point represents an individual player. The horizontal axis indicates the SHAP value, measuring how much a feature increases or decreases the predicted EA FC rating relative to the model's baseline. Positive SHAP values push predictions toward higher ratings, while negative values reduce them.

Several clear patterns emerge from this visualization. First, the market value variables display a strong and consistent relationship with predicted ratings. Players with higher market values generally produce positive SHAP contributions, indicating that increased economic valuation is associated with higher predicted EA FC ratings. Second, the spread of SHAP values reveals that peak market value and previous market value contribute strongly across a wide range of players, reinforcing the notion that both historical peak performance and recent valuation signals are important indicators of player quality. Third, the influence of age-related features appears more complex and dispersed across the SHAP axis, suggesting that age interacts with other factors such as market value and league context, reflecting the inherently non-linear nature of career progression in professional football. Finally, categorical variables such as league and position show

smaller but still noticeable effects, refining predictions rather than determining them directly.

Overall, the SHAP value distribution confirms that market valuation signals dominate the model's predictions, while contextual variables such as league and position play a supporting role in refining final rating estimates.

Structural Analysis of Market Value Impact

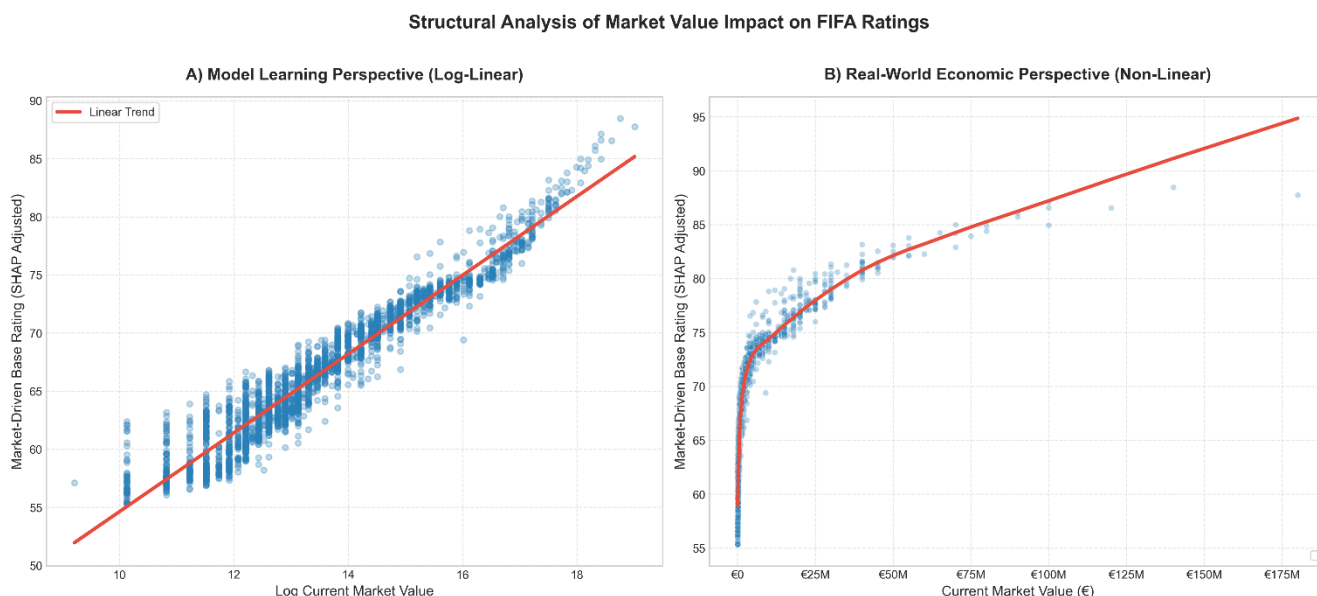


Figure-5

Structural Analysis of Market Value Impact on EA FC Ratings

To further investigate the relationship between economic valuation and player ratings, a structural analysis was performed by isolating the Market-Driven Base Rating. This metric represents the rating component directly attributed to a player's current market value, adjusted using SHAP values to remove the influence of other confounding features (such as age, league, or position).

Figure 5 **presents** this relationship from two distinct perspectives:

A) Model Learning Perspective (Log-Linear)

The left panel (A) illustrates the relationship between the Logarithm of Current Market Value and the adjusted base rating. By applying a logarithmic transformation, we mitigate the extreme skewness of market valuations and reveal a remarkably linear and consistent trend. The high density of data points along the linear trend line suggests that the model's

internal logic follows a stable, proportional rule: every unit increase in log-value corresponds to a steady rise in predicted base rating. This pattern is consistent with the hypothesis that the XGBoost model has captured the underlying "valuation-to-quality" heuristic used in the football industry.

B) Real-World Economic Perspective (Non-Linear)

The right panel (B) displays the same relationship using Actual Market Value (in millions of €). This perspective reveals the real-world economic dynamics:

- **Rapid Growth at Lower Valuations:** In the lower price brackets (€0–25M), small increases in market value correspond to significant jumps in player ratings.
- **Diminishing Returns at the Elite Level:** As players move into the "Elite" category (€75M+), the curve flattens. Large increases in market value result in smaller incremental gains in overall rating.

This non-linear behavior accurately reflects the football market: it is exponentially harder and more expensive to upgrade a player from an 88 to a 91 rating than it is to move a player from 70 to 75.

Key Insight: Together, these visualizations indicate that while the raw market is non-linear and "top-heavy," the underlying relationship between value and player quality is structurally sound. The log-linear alignment in Panel A provides evidence that market signals from Transfermarkt serve as a reliable and organized proxy for professional assessment of player ability.

The Aging Paradox: Experience vs. Investment Value

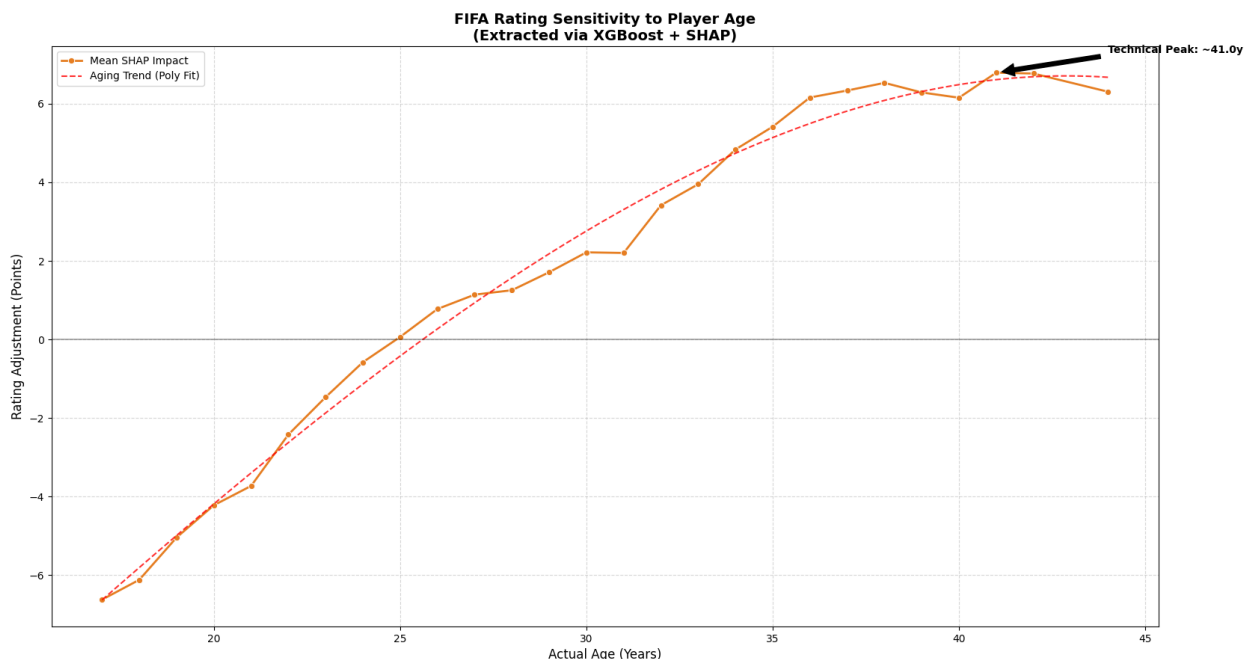


Figure - 6

EA FC Rating Sensitivity to Player Age

To understand how a player's age independently influences EA FC ratings, we extracted the mean SHAP impact for each age year, isolating the contribution of age while controlling for market value and other variables. The resulting curve reveals a consistent upward trend that contrasts with the typical lifecycle of market value in professional football.

Early Career: The Potential Discount (Age 17–25)

At the beginning of a player's career, age acts as a significant negative adjustment to predicted rating. For example, a 17-year-old player receives roughly a –6 rating adjustment from the age feature alone. This penalty gradually diminishes as players mature and establish themselves professionally. However, this does not imply that young players are undervalued in the transfer market; in fact, the opposite is often true. Young players frequently command higher market prices relative to their current performance level because a substantial portion of their valuation represents investment potential. In the football economy:

Market Value = Current Ability + Future Potential

Clubs are effectively purchasing expected future development. This investment component is conceptually similar to the "Potential" attribute in EA FC, which represents possible future growth rather than current ability. Because the model aims to predict current rating, the SHAP analysis reveals a negative age adjustment for very young players, effectively separating current quality from speculative investment value.

Prime and Veteran Phase: The Experience Premium (Age 30+)

As players age, the model assigns increasingly positive rating adjustments, reflecting an experience premium. Around the mid-30s, age contributes positively to predicted rating, capturing factors such as:

- Tactical intelligence
- Professional consistency
- Reputation and legacy within the sport

This trend highlights a key structural difference between football economics and EA FC ratings. While experienced players may maintain elite performance levels, their market value often declines due to limited resale potential. From an economic perspective:

Market Value decreases with age because investment potential disappears.

Older players typically have low future transfer value, meaning clubs cannot expect a profitable resale. Consequently, their transfer market price declines even when current ability (and EA FC rating) remains high.

The Technical Peak

The SHAP curve suggests a technical peak around age 40–41, where the age feature contributes the largest positive adjustment. While only a small number of players remain active at this age, those who do are typically exceptional veterans—often elite goalkeepers or historically great players whose technical consistency and reputation sustain very high ratings.

Key Insight: This analysis **suggests** that Transfermarkt values and EA FC ratings measure fundamentally different concepts:

- **Transfermarkt Market Value:** Current Ability + Investment Potential
- **EA FC Rating:** Current Ability + Reputation / Experience

By modeling these variables with XGBoost and interpreting them using SHAP, we are able to separate the investment component of football economics from the pure performance signal, revealing how age drives the divergence between market value and in-game ratings.

League Impact on EA FC Ratings (Systematic and Local Effects)

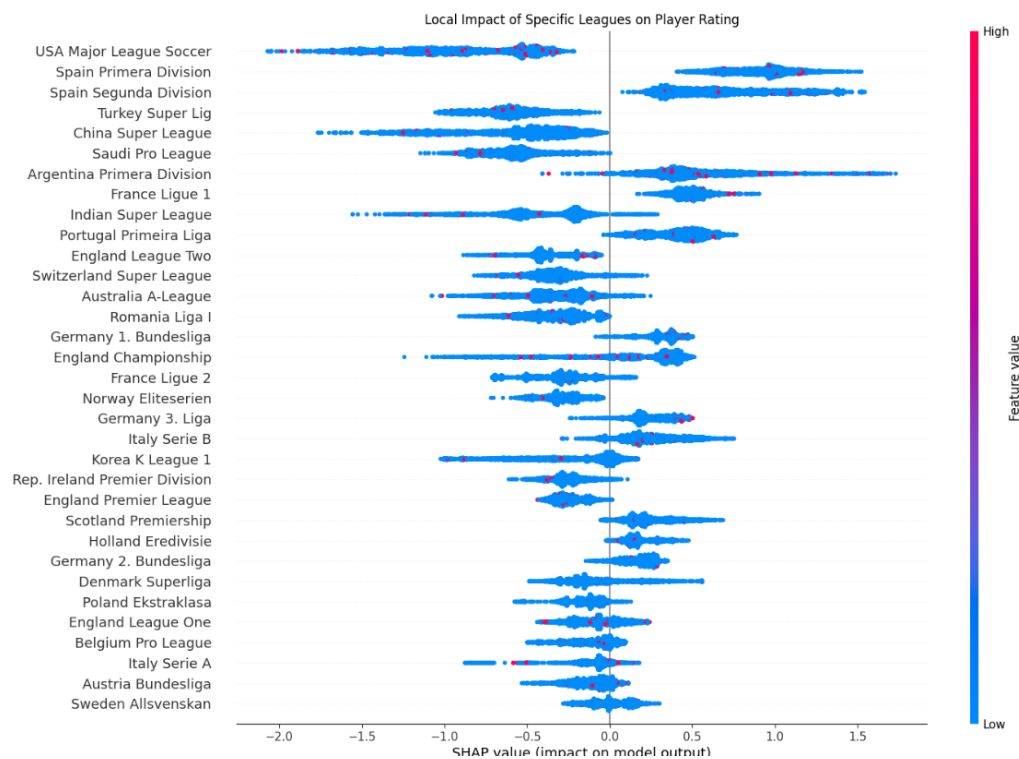


Figure - 7

Systematic and Local Impact of Leagues on Player Rating

The analysis of league-based SHAP values reveals how the player's league acts as a calibrating factor for final rating. Even after accounting for market value and age, the model applies an adjustment between -1.0 and +1.0 points based on the league. This is not random variation; rather, it reflects systematic differences between Transfermarkt market valuation and EA FC rating construction.

1. The "Prestige" Adjustment (The Positive Bar)

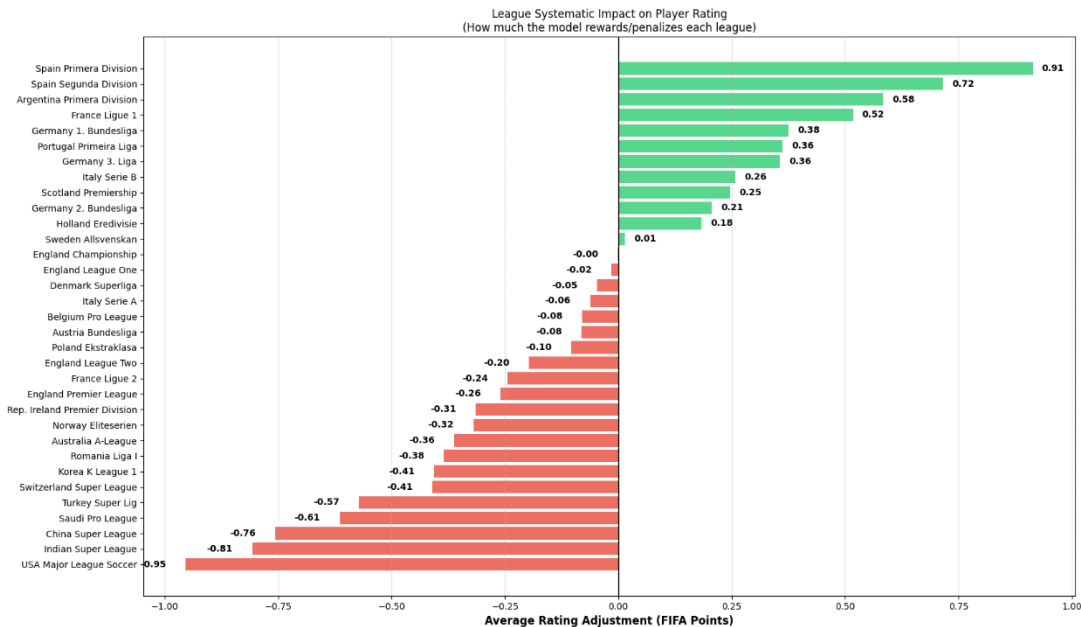


Figure - 8

Leagues such as Spain Primera Division (+0.91), France Ligue 1 (+0.52), and Germany Bundesliga (+0.38) show positive adjustments. In these "Top 5" leagues, a player might have a relatively moderate market value compared to a superstar, but because they play in a high-prestige environment, EA FC often grants them a higher baseline rating.

Transfermarkt prices are highly sensitive to performance and age, but EA FC ratings are often stickier and influenced by league reputation. The model adds these points to "correct" the market value which might not fully capture the prestige of playing in La Liga.

2. The "Market Inflation" Penalty (The Negative Bar)

Leagues such as USA MLS (-0.95), Chinese Super League (-0.76), and Saudi Pro League (-0.61) show significant negative adjustments. In these leagues, players—especially aging stars or "marquee" signings—often have inflated market values or salaries that do not necessarily reflect current technical "Overall" in EA FC terms. Transfermarkt might value a player in MLS or Saudi League highly due to commercial appeal or high transfer fees paid by these leagues. However, EA FC's rating system treats these leagues as lower in technical quality. Therefore, the model subtracts points to bring the market-based prediction down to the realistic EA FC rating.

3. The "Lower Division" Paradox

Interestingly, Spain Segunda Division (+0.72) and Germany 3. Liga (+0.36) show positive adjustments. This suggests that players in these leagues are often "undervalued" in terms

of price-to-rating. They may be very solid players (Rating 65–72) but have very low market values because they are not in the global spotlight. The model adds points to compensate for their low market price.

4. Local Impact (Figure 7 - Beeswarm)

The Beeswarm plot shows variance within each league. For USA Major League Soccer, the long tail toward the left (negative) shows that for the majority of players, being in MLS "drags down" rating prediction relative to their price. For Argentina Primera Division, the wide spread indicates that the league contains both "undervalued gems" (positive SHAP) and "overpriced prospects" (negative SHAP), requiring individual adjustments.

Tactical Biases: How Position Influences the Price-to-Rating Ratio

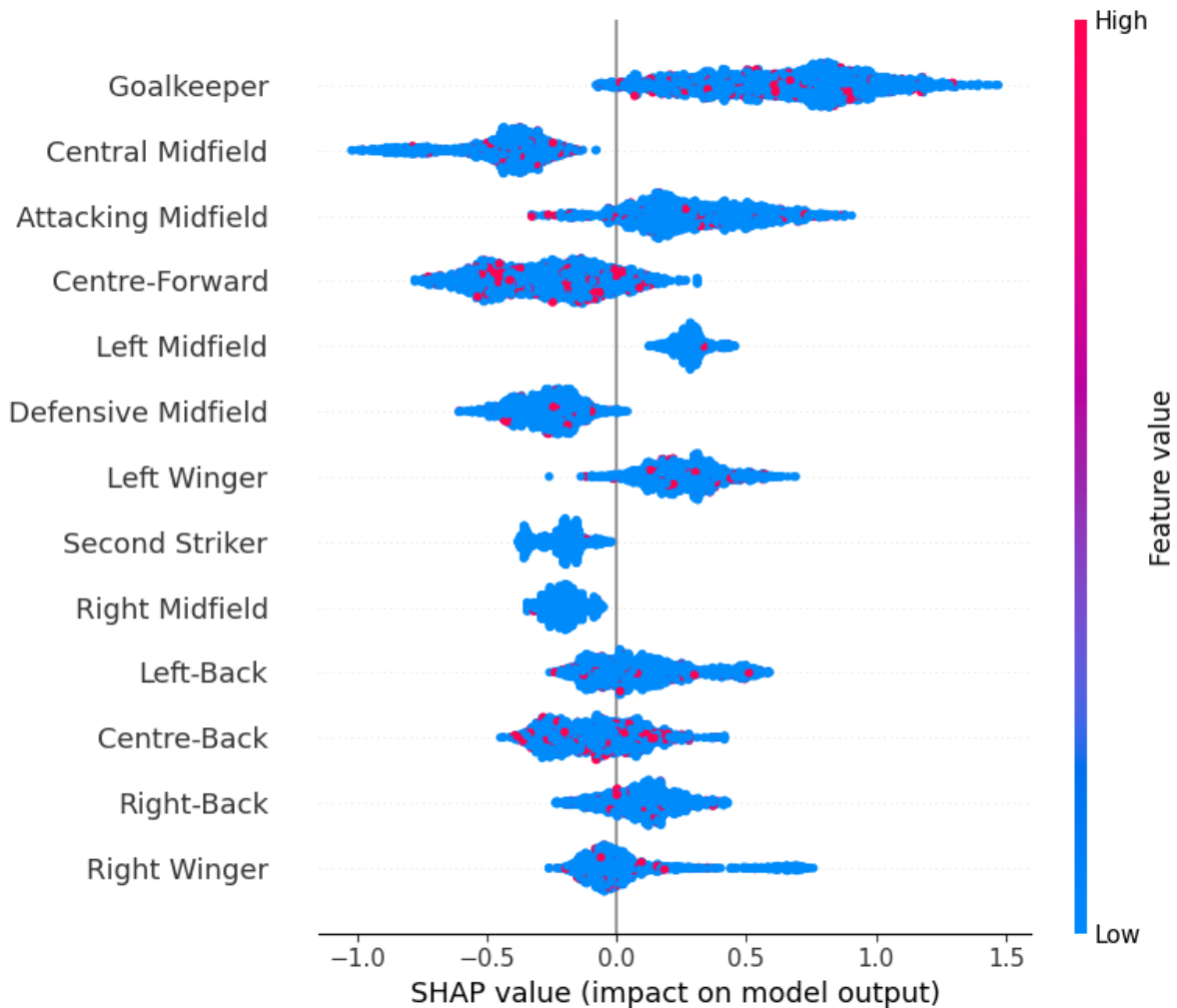


Figure - 9

Positional Impact on EA FC Ratings

The model reveals a systematic adjustment based on a player's tactical role, suggesting that a single euro of market value does not buy the same amount of "Overall Rating" across different positions.

1. The Goalkeeper & Defender "Undervaluation" (Positive SHAP)

Goalkeepers (+0.58) and full-backs (left-back/right-back) show consistent positive adjustments. Historically, goalkeepers and defensive players command lower transfer fees and market values compared to offensive stars, even when world-class in their role. Because their market value is "low" relative to importance, the model adds nearly +0.6

points to goalkeepers to reach their correct EA FC rating. This pattern is consistent with the hypothesis that goalkeepers are "cheaper" per rating point than any other position.

2. The Midfield "Congestion" (Negative SHAP)

Central midfielders (-0.41) and defensive midfielders (-0.26) receive the largest negative adjustments. Central midfield is often the most "expensive" area in terms of market density. High-volume passing and physical presence in the middle of the park are highly valued by Transfermarkt, leading to high prices. However, EA FC's rating formula for central midfielders is often strict. The model must subtract points to ensure that a highly priced midfielder does not receive an unrealistically high rating.

3. The Offensive Premium: "Flashiness vs. Utility"

Attacking midfielders (+0.30) and wingers show positive adjustments, while centre-forwards (-0.26) show negative ones. Top-tier centre-forwards (strikers) have the highest market values in the world. However, unless they are elite (85+), their prices often outpace actual "Overall" rating due to the "Scorer Premium" in the transfer market. Conversely, attacking midfielders often have a "utility" bonus in EA FC. The model rewards them with additional points, possibly because technical stats (dribbling/passing) contribute more heavily to Overall than raw market price suggests.

4. Variance Analysis (Beeswarm Plot)

The Beeswarm plot (local impact) shows that for goalkeepers, impact is almost always positive (the blue cluster is to the right of zero), indicating that the "Goalkeeper Discount" in the market is a universal pattern across the dataset. Conversely, for central midfielders, impact is consistently negative, suggesting a universal "Midfielder Premium" in market pricing.

Conclusion and Final Reflections

The results of this study demonstrate a strong statistical relationship between player market values on Transfermarkt and overall ratings in EA Sports FC. Even a relatively simple baseline model exhibits meaningful alignment between the two systems, suggesting that both may capture similar underlying signals related to player evaluation.

1. Economic Signal vs. Game-Based Ratings

Although high explanatory power is observed ($R^2 \approx 0.90$), it is important to emphasize that Transfermarkt values and EA FC ratings represent fundamentally different conceptual frameworks. Transfermarkt market values reflect economic valuation influenced by

performance, contract conditions, age, and market dynamics, while EA Sports FC ratings are designed as a structured representation of player attributes for gameplay purposes. The XGBoost modeling approach allows estimation of the extent to which variation in EA FC ratings is statistically associated with market-based and contextual features such as age, league, and position.

2. Role of Model Interpretation

SHAP-based analysis provides insights into how different features contribute to model predictions. Market value-related variables, age, and contextual features such as league and position were found to be the most influential factors in explaining variation in EA FC ratings. These results highlight that the model captures consistent statistical relationships between market valuation signals and EA FC ratings, without implying causal relationships between the two systems.

3. Practical Interpretation and Scope

From an applied perspective, the model demonstrates that market-based signals contain substantial information related to EA FC ratings, which may be useful for large-scale analysis of player datasets where EA FC ratings are unavailable. However, it is important to note that the model reflects statistical associations rather than absolute measures of player quality. Predictions should therefore be interpreted as learned patterns from historical data rather than absolute evaluations of player ability.

Overall, the findings suggest that EA FC ratings and Transfermarkt valuations share a significant common statistical signal, despite being designed for fundamentally different purposes.

Final Note

This study provides a data-driven perspective on the relationship between football market valuation and in-game rating systems. While the alignment between the two is strong, the differences in their underlying design objectives should always be considered when interpreting the results.